



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO KPS AGENCY

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

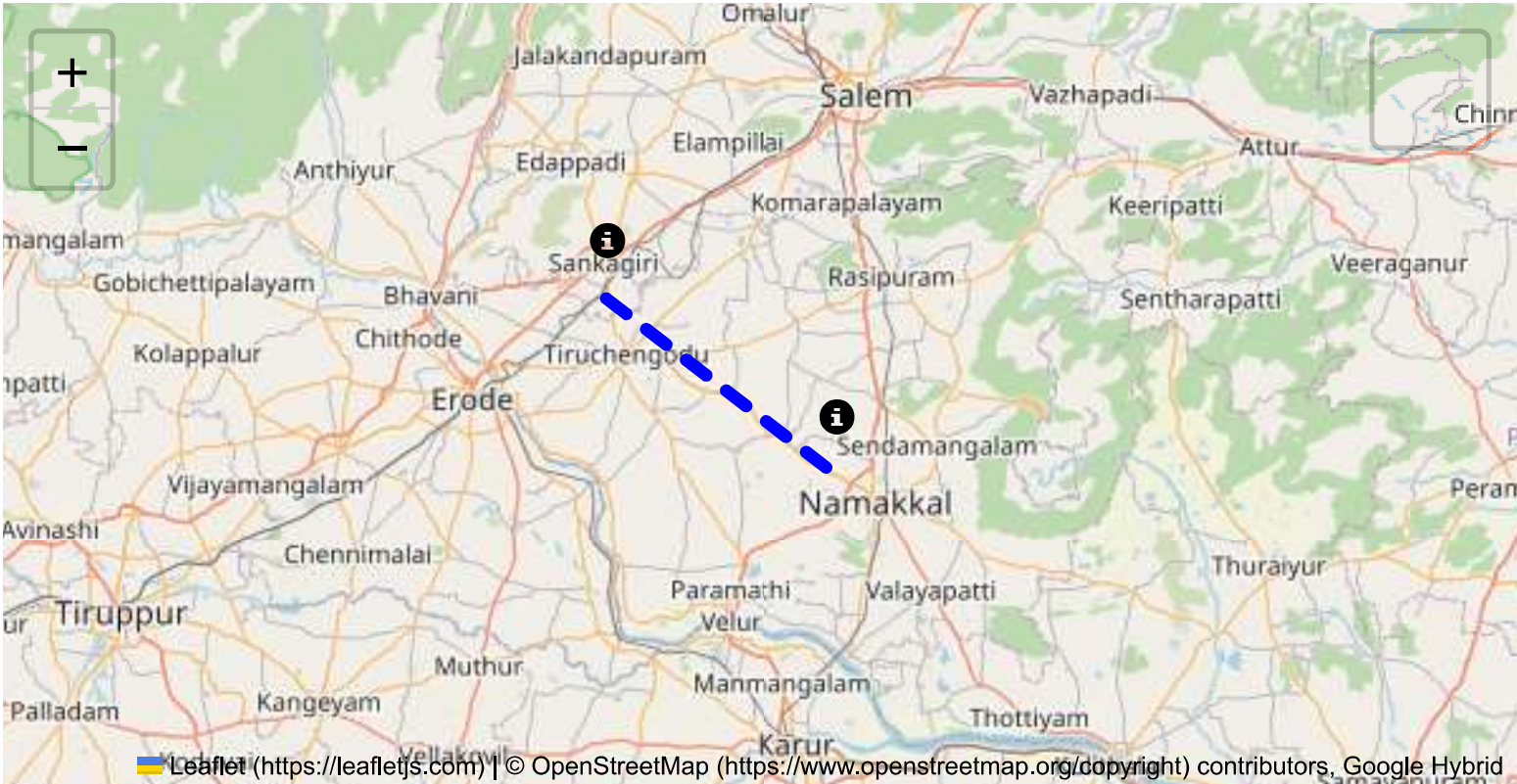
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

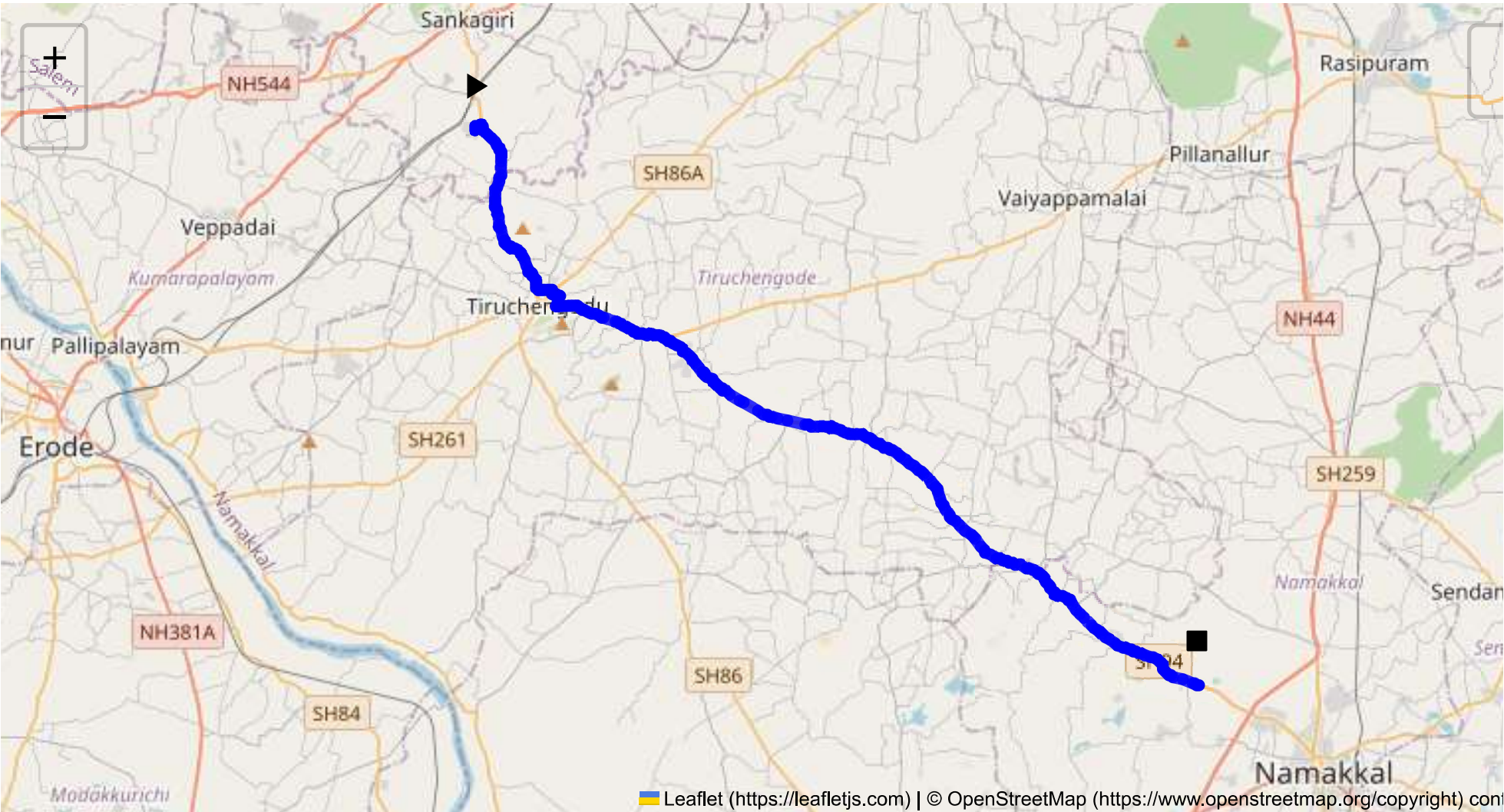
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 38.45 km
Estimated Duration: 0.8 hours
Adjusted Duration (Heavy Vehicle): 1.1 hours
Start: (11.4381, 77.8734)
End: (11.249331, 78.123051)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map: The route from Sangagiri to the SMR Textile Mill in Tiruchengode is approximately 38.45 km and typically takes around 50 minutes for heavy vehicles. This journey will generally involve traversing local roads, passing through both urbanized and rural landscapes of Tamil Nadu. Key towns along the way could include Pallipalayam and minor rural settlements.

2. Typical Weather Conditions and Potential Weather-Related Hazards: The region experiences a tropical climate, with hot summers, moderate winters, and monsoon seasons. Look out for:

- **Summer (March to June):** Dry and hot weather, with temperatures occasionally reaching up to 40°C.
- **Monsoon (July to September):** Heavy rainfall could lead to waterlogging and reduced visibility, affecting driving conditions.
- **Winter (October to February):** Moderate temperatures with occasional fog in the early mornings that can reduce visibility.

3. Analysis of Traffic Patterns: Traffic congestion is common during the following periods:

- **Morning peak hours (7:00 AM - 10:00 AM):** Around towns and key junctions.
- **Evening peak hours (5:00 PM - 8:00 PM):** Increased activity especially in urban areas. Congestion-prone areas include markets and junctions at town centers.

4. Assessment of Road Quality and Infrastructure:

- **State Highways and Major Roads:** Generally well-maintained but can have uneven sections.
- **Local Roads:** May vary in quality; some rural sections could lack proper shoulders and have potholes.

5. Suggestions for Alternative Routes for Emergencies:

- **Via SH 86:** Alternative routes might take longer but could bypass crowded towns.
- For emergencies, use bypass roads around towns to avoid congestion.

6. Summary of Local Regulations Affecting Hazardous Material Transport:

- Tamil Nadu enforces strict regulations regarding the transport of hazardous materials, requiring permits and adherence to specific routes.
- Restrictions on carrying hazardous materials during peak hours in urban areas are common.

7. Overview of Historical Incidents:

- Historical data suggests occasional accidents related to road quality or monsoon weather.
- Greater vigilance is required during monsoon season due to slippery roads and potential landslides or flooding.

8. Environmental Considerations and Sensitive Areas:

- Routes pass through agricultural lands; utmost care should be taken to avoid spillage which can impact the environment.
- Local wildlife and floral reserves must be respected, particularly in rural stretches.

9. Analysis of Communication Coverage:

- Mobile network coverage is generally good in urban areas but may be weak in rural zones, particularly in low-lying areas or away from main roads.

10. Estimated Emergency Response Times:

- **Urban Areas:** Emergency response times could be within 15-30 minutes due to proximity to services.

- **Rural Areas:** Expect longer response times, often up to an hour, due to lesser accessibility.

12. Overall Summary of Risk Assessment: The route from Sangagiri to the SMR Textile Mill is moderately challenging given the climatic and topographical conditions. The key risks involve weather impacts during the monsoon, road quality in rural sections, and potential traffic congestion in urban areas during peak hours. Transporting hazardous materials requires strict compliance with local regulations, and drivers must remain vigilant about environmental and communication issues that could impede safe travel. Regular route assessments before any trip, especially during monsoon, could mitigate many potential hazards, ensuring a smooth journey.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	U-Turn	High	11.2490908, 78.12459989999999	10 KM/Hr
2	Turn	High	11.43811, 77.87348	15 KM/Hr
3	Turn	Medium	11.43956, 77.87340	30 KM/Hr
4	Turn	High	11.43968, 77.87345	15 KM/Hr
5	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
6	Turn	High	11.38346, 77.89975	15 KM/Hr
7	Turn	Medium	11.38190, 77.90254	30 KM/Hr
8	Turn	High	11.38186, 77.90258	15 KM/Hr
9	Blind Spot	Blind Spot	11.37841, 77.90140	10 KM/Hr
10	Blind Spot	Blind Spot	11.24909, 78.12460	10 KM/Hr
11	Turn	High	11.24902, 78.12457	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
5	hospital	Jayaa Hospital	11.3848223, 77.9021886	30 km/h	Medium
8	hospital	Vivekananda Medical Care Hospital	11.364562, 77.9482668	30 km/h	Medium
10	hospital	Manickampalayam, Government Hospital	11.32887, 78.017713	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
6	college	Vivekanandha Educational Institutions for Women	11.3596633, 77.9435556	30 km/h	Medium
7	college	Vivekanandha Dental College	11.3643705, 77.9475897	30 km/h	Medium
9	college	Vivekanandha College of engineering for Women	11.3565938, 77.9465342	30 km/h	Medium
11	college	The Surabi College of Polytechnic	11.2660726, 78.0939552	30 km/h	Medium
12	college	CMS College Of Engineering	11.2647508, 78.0969044	30 km/h	Medium
13	school	Porasapalayam Government School	11.2504918, 78.1163024	30 km/h	Medium
14	school	Sri Vinayaga Higher Secondary School	11.2479837, 78.1182908	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.38395, 77.89480



Risk Type: U-Turn
Risk Level: High
Speed Limit: 10 KM/Hr
Coordinates: 11.2490908, 78.12459989999999



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38346, 77.89975



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.38190, 77.90254



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.38186, 77.90258



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.37841, 77.90140



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.24909, 78.12460



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.24902, 78.12457
